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Adaptive graph convolutional network-based short-term passenger flow prediction for metro

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ABSTRACT

With the development and acceleration of urbanization, urban metro traffic is gradually growing up to a large network, and the structure of topology between stations becomes more complex, which makes it increasingly difficult to capture the spatial dependency. The vertical and horizontal interlacing of multiple lines makes the stations distributed topologically, and the traditional graph convolution is implemented on the adjacency matrix generated based on a priori knowledge, which cannot reflect the actual spatial dependence between stations. To address these problems, this paper proposes an adaptive graph convolutional network model (Adapt-GCN), which replaces the fixed adjacency matrix obtained from a priori knowledge in the traditional GCN with a trainable adaptive adjacency matrix. This can not only effectively adjust the weights of correlations between adjacent stations, but also adaptively capture the spatial dependencies between non-adjacent stations. This paper uses the Shanghai Metro dataset to verify the effectiveness of the model in improving prediction accuracy and reducing training time.

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Deep learning; graph convolutional network; metro card swiping data; short-term passenger flow prediction; urban traffic

Introduction

With the rapid development of domestic intelligent transportation system and metro traffic industry, the smart IC card has become popular, thus generating a large amount of card data. Based on this, an accurate and effective passenger flow prediction model can be built to further perceive the changing trend of passenger flow and provide a powerful decision basis for metro traffic managers, as well as to better plan a smoother travel route and choose a more suitable travel time for passengers, thus avoiding or alleviating the congestion of metro traffic. Therefore, it is of great theoretical significance and practical value to study the problem of passenger flow prediction of metro traffic.

There has been a lot of research work on the construction of passenger flow forecasting models in metro traffic, which can be broadly classified into three categories of passenger flow forecasting models for metro traffic. The first type of models is based on mathematical statistics, such as Auto-Regressive Integrated Moving Average (ARIMA) model (Yu et al., 2021; Wu et al., 2014; Chandra & Al-Deek, 2009). Though the calculation is simple and fast, it

relies on the assumption of smooth distribution of time series, cannot express traffic flow data with highly nonlinear characteristics. The ARIMA model was used to predict the daily passenger flow of a single metro station under normal conditions, and a combination of time series and regression analysis to predict the passenger flow under abnormal conditions with high accuracy (Bai, 2017). However, it only predicts the passenger flow for a single station, which leads to the fact that the only factor that can be considered is the time factor and does not involve the spatial dependence between stations.

The second category is a nonparametric-based intelligent prediction model, Such as Long Short-Term Memory (LSTM) model (Hochreiter & Schmidhuber, 1997). A long and short-term neural network structure is proposed, to capture nonlinear traffic dynamics in an efficient manner, which can overcome the error decay problem of back propagation through memory blocks, thus exhibiting superior capability for time series prediction with long time dependence (Ma et al., 2015). A fused attention mechanism bidirectional long and short-term memory model is proposed, which consists of capturing bidirectional temporal and spatial feature

dependencies from data through bidirectional LSTM (BiLSTM) and attention mechanism units, and through BiLSTM layers. The model considers both forward and backward dependencies of time series data (Li et al., 2022). A spatio-temporal long and short-term memory (ST-LSTM) network is constructed, which achieves the prediction of outbound passenger traffic at multiple stations based on the time cost matrix and the spatial cost matrix (Tang et al., 2019). The models both consider the influence of spatial features between stations and achieve simultaneous prediction of passenger flow at multiple stations. However, it only predicts the passenger flow in one direction for inbound or outbound stations, and does not realize the simultaneous prediction of inbound and outbound passenger flow, which is very detrimental to the metro's transportation management department of the stations for targeted security deployment.

The third category is a hybrid model based on multi-feature extraction. The work used a convolutional neural network (CNN) to learn spatial features from the image data of metro passenger flow, and also used a bidirectional LSTM to extract temporal features from the time series of metro passenger flow, and finally fused the spatio-temporal features by a fully connected network, and then obtained the prediction results of passenger flow (Ma et al., 2019). However, this method needs to grid the study area first when describing the image data of metro passenger flow. The interval size of its divided grid is difficult to determine; if the interval is divided too large, a grid may contain multiple stations; if the divided interval is too small, many useless grids will exist, resulting in the waste of computational resources. For the station-level passenger flow prediction problem of metro traffic, a graph convolutional network (GCN) is constructed using deep learning methods to capture the spatio-temporal features of passenger flow changes (Han et al., 2019). The work mines spatial features between metro stations with ResNet, extracts topological information in the metro network with GCN, and extracts temporal features with LSTM and incorporates the effects of factors such as weather conditions and air quality (Zhang et al., 2021). Although the inclusion of multi-source heterogeneous information such as weather and air conditions can bring some accuracy improvement to the model, the information is difficult to collect and process, and the data from multiple sources can lead to low model efficiency due to the redundancy of information and overly complex structure in the model.

In summary, the current passenger flow forecasting problem of metro traffic mainly lies in inadequate mining of spatial-temporal features. The mining of temporal features is mostly based on typical time series models, such as ARIMA; while in terms of spatial features mining, the models used for different metro traffic scenarios have great variability, and GCN can better extract the spatial topology information for multi-line metro traffic network. There has been some work related to adaptive graph neural networks. For example, an adaptive graph neural network applied to the field of chemistry, and an adaptive heterogeneous graph neural network for V2X application (Li et al., 2018; Xu et al., 2022). In this paper, we focus on the spatial feature mining of passenger flow changes in multi-route metro traffic, and the research idea is to achieve simultaneous prediction of passenger flow in and out of multiple stations based on the improved GCN model, and the model is required to be as simple and easy to operate as possible.

Overall, the contributions of our work are as follows:

- We study the metro passenger flow prediction problem and propose a new graph convolution module, namely Adaptive Graph Convolution Neural Network (Adapt-GCN), to capture spatial correlations between adjacent and non-adjacent stations using adaptation.
- The fixed adjacency matrix in traditional GCN is replaced with a trainable adaptive adjacency matrix, and short-time passenger flow prediction of metro traffic is achieved by stacking multiple Adapt-GCN layers and adding a residual network, and finally connecting a fusion layer.
- We have conducted extensive experiments on real-world datasets. The experimental results show that our model consistently outperforms all baseline methods.

Problem description

The passenger flow prediction problem for urban metro traffic is a typical spatio-temporal sequence prediction problem, which can use the observation results of the first r time steps to predict the incoming and outgoing passenger flows of all stations in future time steps. In this paper, a city-wide metro network structure is defined on a graph, and a Graph Convolutional Neural Network (GCN) is used to learn its spatial features. Common spatial features in urban metro traffic include: strong similarity of passenger flow variation among different stations due to the

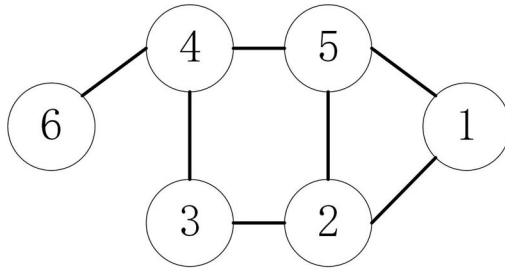


Figure 1. Adjacency matrix (take left legend as an example).

influence of population density and POI attributes in a certain range; distinct interaction of passenger flow direction among different stations due to the influence of popular routes, such as most of the incoming passengers at station a will exit from station b , or outgoing passengers at station a will mainly come from station b (Liu et al., 2022).

From the above, it can be seen that the passenger flow changes among stations have strong similarity and interaction, but in the process of fully mining and applying them, on the one hand, it is difficult to achieve due to the large amount of specialized domain knowledge required; on the other hand, people's thinking perspective when mining features may not be comprehensive. With the rapid development of traffic data collection and storage technologies, more and more researchers have turned their attention to data-driven approaches, and deep learning-based passenger flow prediction models, which advocate feature sensing from data, have in turn been widely recognized by scholars. Among them, GCN is usually used to extract spatial features in metro traffic road networks, and the most prominent metro traffic network graphs G with feature matrices are X defined as follows.

Metro traffic network map

In this work, we define an urban transit network on a graph. The graph of a metro traffic network consists of nodes, edges, and the weights of the edges, usually denoted by a graph $G = (V; E; A)$, where V is the set of all stations, corresponding to the observations from each metro station in the transit network, $V = \{v_1; v_2; \dots; v_n\}$, where n is the number of stations and E is the set of edges, indicating the connectivity between stations. Denote G by $A \in R^{n \times n}$ the adjacency matrix, which contains only 0 and 1 elements, if there is an edge between stations i and j , $A_{i,j} = 1$, otherwise $A_{i,j} = 0$; let $A_{i,i} = 0$. As an example, Figure 1 shows the topology and the adjacency matrix A of a simple network diagram:

0	1	0	0	1	0
1	0	1	0	1	0
0	1	0	1	0	0
0	0	1	0	1	1
1	1	0	1	0	0
0	0	0	1	0	0

From Figure 1, it can be seen that the metro network graph G belongs to a simple undirected graph, which A is an adjacency matrix with only 0 and 1 elements, and the diagonal elements are all 0. For a simple undirected graph with n vertices, the Laplacian matrix $L \in R^{n \times n}$ is defined as (Shuman et al., 2013)

$$L = D - A \quad (1)$$

Where A is the adjacency matrix and A_{ij} denotes the weights of the edges from node i to node, the degree matrix D is a diagonal matrix, consisting of node degrees, $D_{ii} = \sum_j A_{ij}$. To normalize the Laplacian matrix in the above equation L .

$$L^{sys} = D^{-\frac{1}{2}}(D - A)D^{-\frac{1}{2}} = I_n - D^{-\frac{1}{2}}AD^{-\frac{1}{2}} \quad (2)$$

Where $I_n \in R^{n \times n}$ is a unit matrix. L^{sys} represents the normalized Laplacian matrix, which is subsequently abbreviated as L uniformly, the Laplacian matrices presented later are normalized Laplacian matrices.

Feature matrix

In this paper, the time-series data of passenger flow is used as the attribute feature of each station (node) in the metro network, denoted as $X \in R^{n \times r \times 2}$, which represents n the number of stations, r represents the number of attribute features, i.e., the window length of the time series, and 2 denotes both inbound and outbound passenger flows. The feature matrix of the input model is expressed as $X \in R^{n \times r \times 2}$, as shown in Figure 2. In the figure, T_r represents the observed passenger flow at m stations at the r th time step, S_m represents the m th station, and in and out represent the inbound and outbound passenger flows, respectively.

The model

Traditional graph convolutional network (GCN)

Traditional GCNs are defined in two ways: the first is the spectral approach (Bruna et al., 2014; Defferrard et al., 2016; Xu et al., 2019), where the convolution is

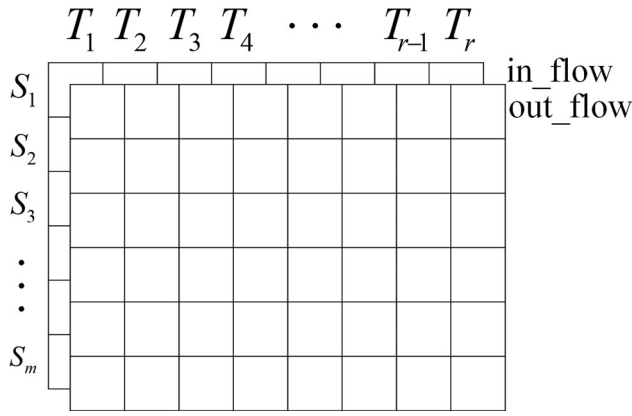


Figure 2. Metro stations passenger flow data into the feature matrix.

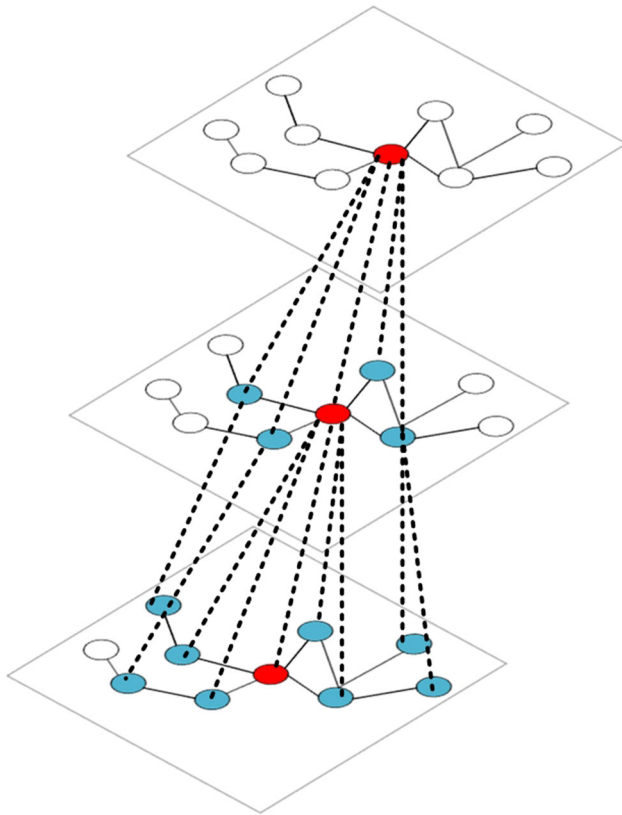


Figure 3. The changes of Receptive field of GCN.

defined by the Fourier transform of the graph and the convolution theorem; the other is the spatial approach (Niepert et al., 2016; Hamilton et al., 2017; Kipf & Welling, 2017), where the convolution is defined as a weighted average function over all vertices located in the neighborhood of the target vertex. The original graph convolution defined the convolution kernel in the frequency domain, solving the problem that the number of neighborhood nodes may be different for different vertices in the spatial domain. Subsequent studies made the graph convolution more effective,

such as Defferrard et al. (2016), who parameterized the convolution filter using polynomial approximation to achieve locality in the spatial domain and used Chebyshev polynomials instead of convolution kernels to achieve better applications, and obtained the graph convolution formula as:

$$y_{out} = \sum_{k=0}^{K-1} \beta_k T_k(\tilde{L})x, \quad \tilde{L} = \frac{2L}{\lambda_{max}} - I_n \quad (3)$$

where, y_{out} denotes the output of the graph convolution, $T_k(\cdot)$ denotes the k -order Chebyshev polynomial, and $K - 1$ denotes the order of the polynomial. k represents the perceptual field of the convolution kernel, i.e., each convolution is a weighted sum of eigenvalues from the k -hop neighbors of the central vertex, and β_k is the weight coefficient, i.e., the learnable parameter in the neural network. $\tilde{L}$ represents the rescaling of the normalized Laplace matrix L . The reason for this rescaling transformation is that the input to the Chebyshev polynomial needs to be between $[-1, 1]$.

In this study, we use a more simplified version of the GCN (Kipf & Welling, 2017). They employ some techniques to further reduce the computational effort of graph convolution. First, a first-order approximation is used to set each graph convolution layer to handle only one-hop neighbors features ($K = 2$), and then multiple graph convolution layers are superimposed by the rule of hierarchical propagation to achieve the effect of multi-hop neighbors feature propagation. The effect of the change of its perceptual field is shown in the following Figure 3.

Subsequently, we further assume $\lambda_{max} \approx 2$, and simplify the parameters, in order to constrain the parameters and stabilize the numerical performance, the two parameters β_0, β_1 are replaced by a single parameter β such that $\beta = \beta_0 = -\beta_1$. Finally, the final graph convolution formula is obtained after rescaling and dimensional generalization as follows:

$$H^{(1)} = \sigma(\hat{A}XW^{(0)}), \quad \hat{A} = \tilde{D}^{-\frac{1}{2}}\tilde{A}\tilde{D}^{-\frac{1}{2}} \quad (4)$$

The above equation generalizes the graph convolution operation to the graph signal $X \in R^{n \times c_{in}}$, where c_{in} denotes the number of channels of the input sequence, i.e., the number of attribute features. In addition, $\tilde{A} = A + I_n$, $I_n \in R^{n \times n}$ is a unit matrix, and $\tilde{D} = \sum_j \tilde{A}_{ij}$, $W \in R^{c_{in} \times c_{out}}$ denotes the parameter matrix of feature variations, i.e., the parameters that can be learned in the neural network. c_{out} represents the number of channels in the output sequence, and an activation function $\sigma(\cdot)$ is applied outside. We get $H \in R^{n \times c_{out}}$, which denotes the feature matrix of the

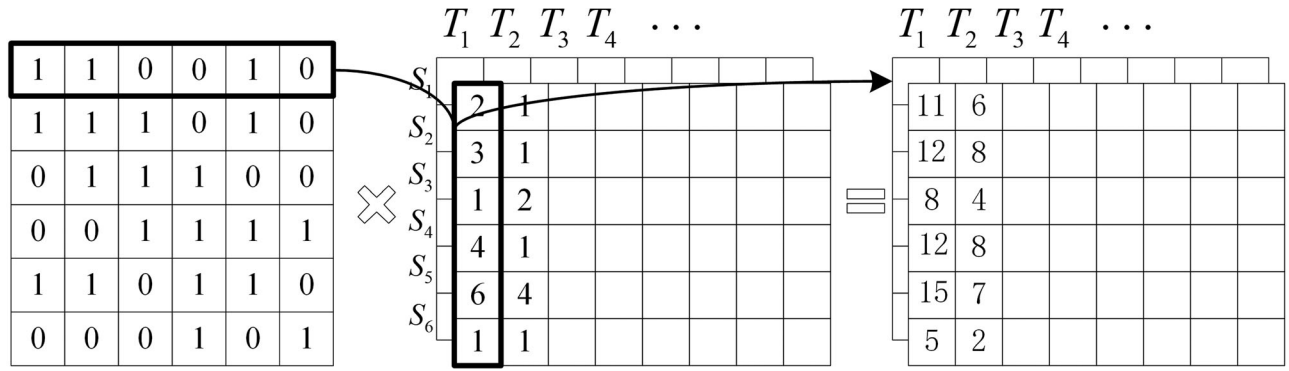


Figure 4. Traditional GCN aggregation node feature information (take Figure 1 as an example).

output after graph convolution, $H^{(1)}$ denotes the output of one layer of graph convolution, and if two layers of graph convolution are stacked, the expression of $H^{(2)}$ is as follows.

$$H^{(2)} = \sigma(\hat{A}\sigma(\hat{A}XW^{(0)}))W^{(1)} = \sigma(\hat{A}H^{(1)}W^{(1)}) \quad (5)$$

Adaptive graph convolutional network

Equation 4 of the traditional GCN calculation can be simplified and expressed as $H = \tilde{A}XW$, where $\tilde{A} = A + I_N$, the activation function and the normalization of the adjacency matrix are omitted here. From the simplified formula, the traditional GCN splits the convolution process into two steps, the first step is to aggregate the feature information of itself and its neighboring nodes by $\tilde{A}X$, where $\tilde{A}$ can be considered as the adjacency matrix with the value range of 0 or 1. This aggregation is simply a summation of the feature information of itself and its neighboring nodes. The second step is the feature variation by W , i.e., mining the nonlinear features among nodes, and W can therefore be called the feature variation matrix. Taking the simple topological network in Figure 1 as an example, the following figure shows the aggregation process of $\tilde{A}X$ in the traditional GCN formulation, i.e., the simple summation of the feature information of itself and its neighbors.

As shown in Figure 4, the adjacency matrix $\tilde{A}$ of traditional GCN is fixed and remains constant throughout the training process of the neural network. However, this paper argues that the fixed adjacency matrix $\tilde{A}$ makes the GCN very limited in aggregating node features, which can neither control the degree of influence that itself and neighboring nodes on the result, nor learn the feature information among non-neighboring nodes other than itself and its neighboring nodes. In this paper, a randomly initialized adjacency matrix is set to learn and optimize through the

training of the neural network, thus construct a more adapted adjacency matrix for the data itself, which does not require a priori knowledge and aims to model adaptively capture hidden spatial correlations (Wu et al. 2019; Song et al., 2020). The GCN model optimized by this method will be referred to as the adaptive graph convolutional network (referred to as Adapt-GCN). The computational formula is as follows.

$$H = \sigma(SXW), \quad S \in R^{n \times n} \quad (6)$$

Where S denotes the randomly initialized adjacency matrix in Adapt-GCN; n denotes the number of nodes; and $X \in R^{n \times c_{in}}$ denotes the feature matrix. Continue to keep the feature change matrix $W \in R^{c_{in} \times c_{out}}$, where c_{in}, c_{out} denotes the number of channels of the input and output sequences, respectively; $H \in R^{n \times c_{out}}$ denotes the feature matrix output by the graph convolution operation. Continuing with the simple topological network in Figure 1, Adapt-GCN aggregates the node feature information in the following way.

Figure 5 shows the aggregation process of SX in the Adapt-GCN formulation. From the illustration, it can be seen that the adjacency matrix S in Adapt-GCN is no longer a fixed adjacency matrix, but a random initialization matrix with the same shape as the adjacency matrix. Through the training of the neural network, it can not only learn the correlations between neighboring nodes more freely, but also the correlations between non-neighboring nodes can be obtained by learning. By stacking multiple Adapt-GCN layers and adding the residual structure, and finally using a fusion layer, the short-time prediction model of multi-station inbound and outbound passenger flow in the metro traffic network in this section, called Adapt-GCN, is constituted with the following structure (Figure 6).

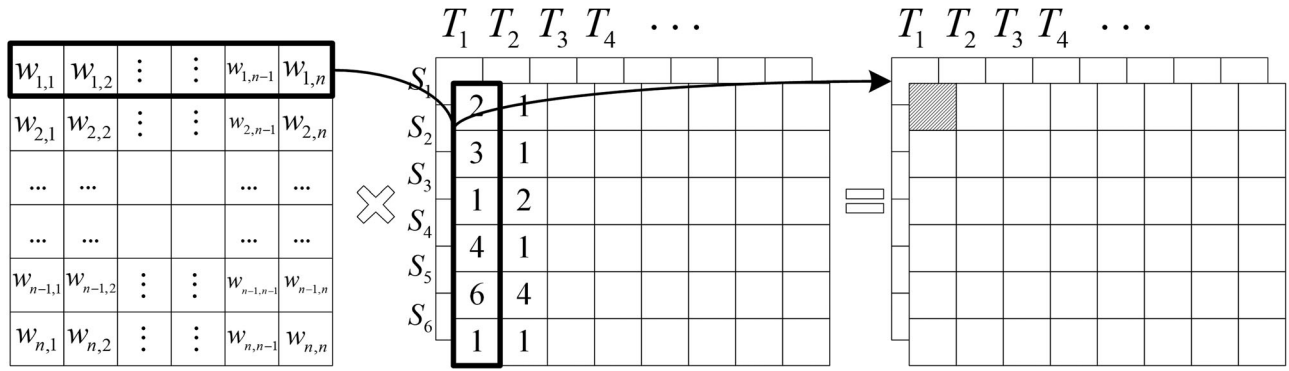


Figure 5. Adapt-GCN aggregation node feature information.

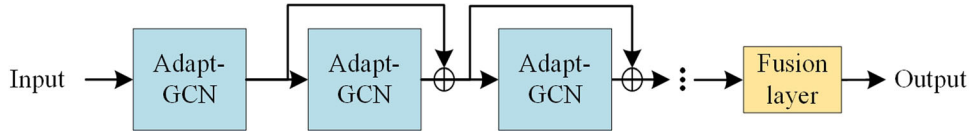


Figure 6. Structure diagrams of Adapt-GCN.



Figure 7. Distribution map of Shanghai metro stations.

Model estimation and empirical analysis

Computational experiments were performed in order to evaluate the performance of the proposed model and solution. The proposed method is coded in Python language. All the experiments are performed on a server with one GPU (Quadro P2000), four CPUs (Intel(R) Xeon(R) W-2102 CPU @ 2.90 GHz).

Data set partitioning

The passenger flow data were obtained using the Shanghai Metro dataset (Liu et al., 2022), which includes 288 metro stations with the topological distribution among stations as shown in Figure 7. The time

range of the data is from July 1st to September 30st, 2016, focusing only on the prediction of incoming and outgoing passenger flow from 5:30 to 23:45 each day. For each station, the inflow and outflow of passengers were counted every 15 min, thus, a total of 73 time steps were generated for one day and a total of 6716 segments of data for 92 days. All data were divided into training set, validation set and test set with the division ratio of 62, 9 and 21 days.

Figure 8 shows the outbound traffic of the same station during a week. It can be seen that weekdays exhibit the same traffic pattern, and the traffic pattern differs significantly between weekends and weekdays.

Evaluation metrics and baseline model

To judge the prediction effect of the model, the mean absolute error (MAE), root mean square error (RMSE) were used to quantify the goodness of the prediction results, and the error was calculated as follows.

$$MAE = \frac{1}{n} \sum_{i=1}^n |\text{observed}_i - \text{predicted}_i| \quad (7)$$

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (\text{observed}_i - \text{predicted}_i)^2} \quad (8)$$

where, observed_i is the true traffic value of the i th test sample, predicted_i is the predicted traffic value of the i th test sample, and n is the total number of test samples. To verify the effectiveness of the model, four classical models (STGCN, ResNet, ResGCN, JK-ResGCN) are selected for performance comparison with Adapt-GCN (Yu et al., 2017; Zhang et al., 2017; Li et al., 2017).

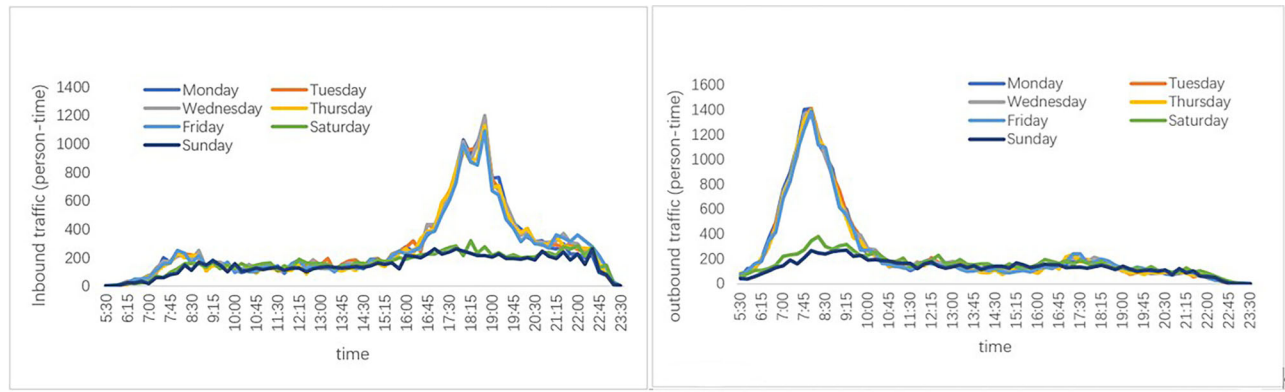


Figure 8. Distribution of passenger flow inbound and outbound on weekdays and weekends by time of day.

Table 1. Experiment results of models.

Models	Inbound traffic (person-time)		Outbound traffic (person-time)		Total traffic (person-time)		Time (sec)
	MAE	RMSE	MAE	RMSE	MAE	RMSE	
STGCN	24.078	44.738	22.4853	45.530	35.300	64.787	1491.511
ResNet	23.877	43.951	23.428	44.838	35.031	64.273	1035.368
ResGCN	25.273	46.189	24.459	46.421	36.417	67.251	1538.014
JK-ResGCN	24.126	43.677	23.651	47.083	35.288	64.567	1845.622
Adapt-GCN	23.446	43.118	21.986	43.378	34.716	62.687	1110.640

Experimental settings

The parameters of all models are set using the principle of tuning and comparison uniformity, where the number of stacking layers of GCN or CNN is 8, the time step of observed data is $r = 5$, MAE and RMSE are loss function and optimizer, respectively, and the learning rate is 0.001. The number of iterations is 50, the batch_size is set to 3, the activation function $\sigma(\cdot)$ is ReLU, and the data is normalized between 0 and 1. The size of the convolutional kernel in the 2D convolutional layer of ResNet is 3×3 , and padding is used to preserve the edge information. The model is constructed in the framework of TensorFlow using the Python language, and GPU acceleration is used in the training process of the model.

Prediction accuracy analysis

In the comparison model, ①STGCN uses a sandwich structure of two 1D time-gated convolutions sandwiched by a graph convolution layer to form a spatio-temporal convolutional block, and builds a deep graph convolutional network by stacking four spatio-temporal convolutional blocks (Yu et al., 2017). Although this model can capture both spatio-temporal features, it takes the feature matrix output from CNN1D as the input of GCN, which leads to inadequate capture of spatial features by GCN compared with the direct mining of spatial features from the original input data. ② The CNN in ResNet is applicable to regular

raster data, which is obviously not applicable to general topological map data (Zhang et al., 2017). ③ResGCN generates “hx0022;over-smoothing”hx0022; in the process of stacking multiple GCNs using residual structure (Xu et al., 2017). ④JK-ResGCN adds a jump knowledge network on top of ResGCN to effectively solve the over-smoothing problem (Li et al., 2017; Xu et al., 2017), but the jump knowledge network puts the output of each GCN layer into the final fusion layer in a stitched manner, and the number of input channels in the fusion layer increases as the GCN layers are stacked, which makes the fusion layer spend more time on feature learning (Table 1).

Effect of the number of GCN layers

By adjusting the number of stacked layers of GCN, the prediction accuracy (RMSE) of the total passenger flow and the training time of the models are compared in detail for the four models of ResGCN, JK-ResGCN, ResNet and Adapt-GCN, respectively. The comparison results of the four models are shown in the following figures.

As shown in Figure 9 above, the prediction accuracy of Adapt-GCN remains optimal regardless of the number of stacked GCN layers, especially in shallow networks. This is due to the presence of an adaptive adjacency matrix S in Adapt-GCN, which allows the model to capture the spatial correlation between distant stations in shallow networks as well. However, as the number of

stacked layers increases, the range of the perceptual field of the other three models is gradually expanded, and the accuracy advantage of Adapt-GCN becomes less and less obvious. In terms of training time complexity, the training time complexity of Adapt-GCN is relatively worse than that of CNN-based ResNet, because CNN can be trained in parallel better. However, compared with ResGCN, which uses a fixed adjacency matrix, the Adapt-GCN feature learning method becomes more flexible, and although the number of learned parameters increases, the overall training time decreases.

Ablation study

To evaluate the effectiveness of the components in the proposed model, we designed ablation experiments and tested a variant of Adapt-GCN on the Shanghai Metro dataset. Specifically, Adapt-GCN w/o residual is

the removal of the residual connection module in the Adapt-GCN model.

Table 2 shows the prediction results of Adapt-GCN after removing the residual connections. It can be seen from the table that Adapt-GCN outperforms Adapt-GCN w/o residual on the Shanghai dataset, which demonstrates the positive effect of residual connectivity on improving the prediction performance.

Adjacency matrix analysis

For the Shanghai Metro dataset, the physical map describing the connectivity between stations is visualized separately from the adaptive adjacency matrix S obtained by the single-layer Adapt-GCN training. The comparison results are shown in the following figure.

Figure 10(a) shows a physical map of size 288×288 in the Shanghai Metro dataset, i.e., the fixed adjacency matrix A , where the dark area values are 1 and the light area values are 0. The values of different areas indicate the connectivity relationships between different stations, respectively. Figure 10(b) shows the adaptive adjacency matrix S obtained by neural network training, where the darker color in the figure indicates the stronger correlation between two stations and the lighter color indicates the weaker correlation. The adaptive adjacency matrix S and the fixed adjacency matrix A , both of which have obvious diagonal lines, represent the strong correlation between neighboring stations. In addition, the adaptive adjacency matrix S becomes a wider range of values compared to the fixed adjacency matrix A , which has only 0 and 1, indicating that it can learn correlations between stations more flexibly. The light-colored area in Figure 10(a) corresponds to the non-zero element in Figure 10(b), which indicates that the single-layer Adapt-GCN can also capture the spatial correlation between distant stations.

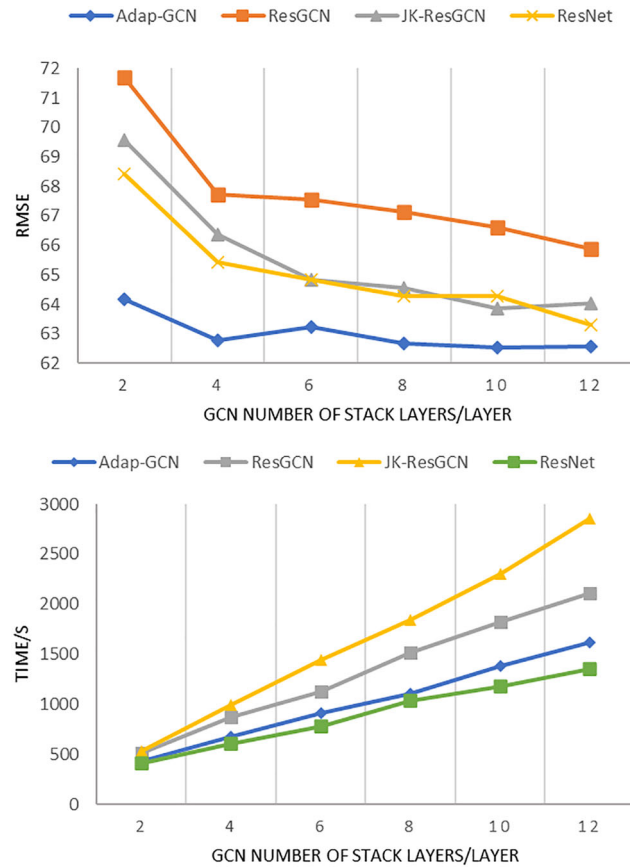


Figure 9. Comparison of accuracy and training time of ResGCN, JK-ResGCN, ResNet and Adapt-GCN.

Table 2. Ablation study performance.

Models	Inbound traffic (person-time)		Outbound traffic (person-time)		Total traffic (person-time)		
	MAE	RMSE	MAE	RMSE	MAE	RMSE	Time (sec)
Adapt-GCN w/o residual	24.212	44.119	22.677	43.835	35.623	64.287	1494.912
Adapt-GCN	23.446	43.118	21.986	43.378	34.716	62.687	1110.640

Conclusion and future work

In this paper, we propose a new improved GCN network based on GCN for passenger flow prediction, for the short-time prediction problem of incoming and outgoing passenger flow of multiple metro stations in a multi-route metro traffic network,

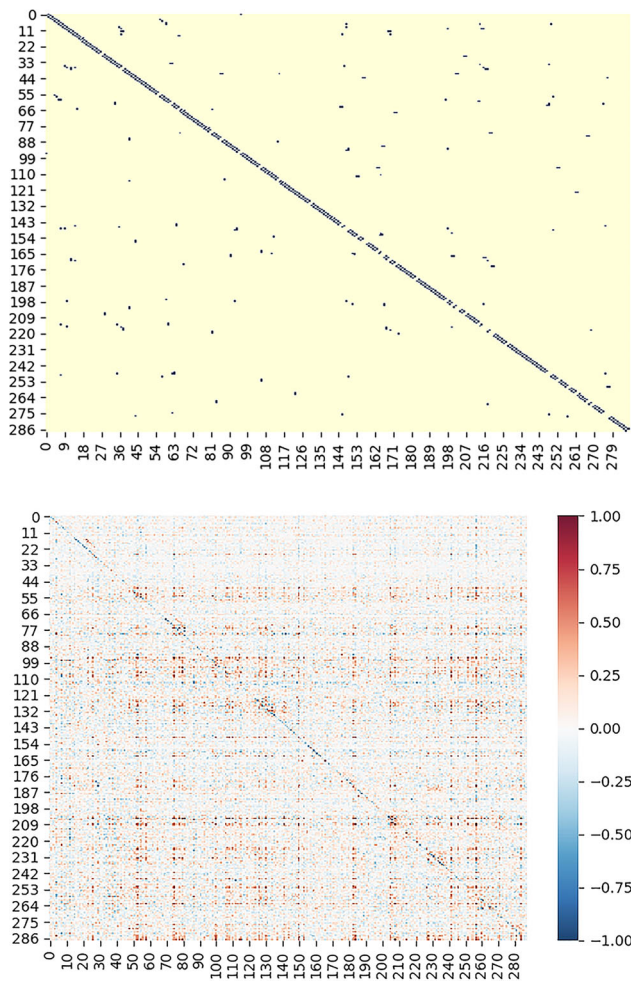


Figure 10. Comparison of fixed adjacency matrix and adaptive adjacency matrix.

considering the spatial distribution between its stations in a topological structure, and replacing the fixed adjacency matrix in the traditional GCN with a random initialization matrix of the same shape, which is optimized by the back-propagation process of the neural network. This approach can adequately capture the spatial correlation between distant stations because it is no longer constrained by the fixed adjacency matrix, but allows the GCN model to adaptively learn the topological relationships between stations from the data. Experiments show that the model outperforms other existing methods on real datasets. On the other hand, the random initialization matrix no longer requires the real topological relationships between stations, but only the total number of stations, which largely alleviates the preparation of the adjacency matrix. Although the proposed model outperforms other methods, the impact of the difference in passenger flow patterns between weekdays and weekends on the accuracy of passenger flow prediction is not considered. In future work, we will pay more attention to

the mining of temporal correlations and consider temporal periodicity, e.g., the morning and evening peaks of each weekday are usually similar, and the passenger flow patterns of each weekend are also similar. We will consider the time factor to further improve the prediction accuracy.

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Disclosure statement

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References

- Bai, L. (2017). Urban rail transit normal and abnormal short-term passenger flow forecasting method. *Journal of Transportation Systems Engineering and Information Technology*, 17(1), 127–135. <https://doi.org/10.16097/j.cnki.1009-6744.2017.01.019>
- Bruna, J., Zaremba, W., Szlam, A., & Lecun, Y. (2014, April 14–16). *Spectral networks and locally connected networks on graphs* [Paper presentation]. 2nd International Conference on Learning Representations, [Paper presentation], Banff, AB, Canada.
- Chandra, S. R., & Al-Deek, H. (2009). Predictions of free-way traffic speeds and volumes using vector autoregressive models. *Journal of Intelligent Transportation Systems*, 13(2), 53–72. <https://doi.org/10.1080/15472450902858368>
- Defferrard, M., Bresson, X., & Vandergheynst, P. (2016, December 5–10). *Convolutional neural networks on graphs with fast localized spectral filtering*. [Paper presentation]. NIPS, Barcelona, ES, Spain.
- Hochreiter, S., & Schmidhuber, J. (1997). Long short-term memory. *Neural Computation*, 9(8), 1735–1780. <https://doi.org/10.1162/neco.1997.9.8.1735>
- Han, Y., Wang, S., Ren, Y., Wang, C., Gao, P., & Chen, G. (2019). Predicting station-level short-term passenger flow in a citywide metro network using spatiotemporal graph convolutional neural networks. *ISPRS International Journal of Geo-Information*, 8(6), 243–267. <https://doi.org/10.3390/ijgi8060243>
- Hamilton, W., Ying, Z., & Leskovec, J. (2017, December 4–9). *Inductive representation learning on large graphs* [Paper presentation]. International Conference on Neural Information Processing Systems, Long Beach, CA, USA.

- Kipf, T. N., & Welling, M. (2017, February 24–26). *Semi-supervised classification with graph convolutional networks* [Paper presentation]. International Conference on Learning Representations, Toulon, Val, France.
- Li, G., Muller, M., Thabet, A., & Ghanem, B. (2019, October 20–26). *Deepgcns: Can gcns go as deep as cnns?* [Paper presentation]. In Proceedings of the IEEE/CVF international conference on computer vision, Seoul, KR, Korea.
- Liu, L., Chen, J., Wu, H., Zhen, J., Li, G., & Lin, L. (2022). Physical-virtual collaboration modeling for intra-and inter-station metro ridership prediction. *IEEE Transactions on Intelligent Transportation Systems*, 23(4), 3377–3391. <https://doi.org/10.1109/TITS.2020.3036057>
- Li, R., Wang, S., Zhu, F., & Huang, J. (2018). Adaptive graph convolutional neural networks. *Proceedings of the AAAI Conference on Artificial Intelligence*, 32(1), 3546–3553. <https://doi.org/10.1609/aaai.v32i1.11691>
- Li, Z., Xu, H., Gao, X., Wang, Z., & Xu, W. (2022). Fusion attention mechanism bidirectional LSTM for short-term traffic flow prediction. *Journal of Intelligent Transportation Systems*, 0(0), 1–14. <https://doi.org/10.1080/15472450.2022.2142049>
- Ma, X., Tao, Z., Wang, Y., Yu, H., & Wang, Y. (2015). Long short-term memory neural network for traffic speed prediction using remote microwave sensor data. *Transportation Research Part C: Emerging Technologies*, 54, 187–197. <https://doi.org/10.1016/j.trc.2015.03.014>
- Ma, X., Zhang, J., Du, B., Ding, C., & Sun, L. (2019). Parallel architecture of convolutional bi-directional LSTM neural networks for network-wide metro ridership prediction. *IEEE Transactions on Intelligent Transportation Systems*, 20(6), 2278–2288. <https://doi.org/10.1109/TITS.2018.2867042>
- Niepert, M., Ahmed, M., & Kutzkov, K. (2016, June 19–24). *Learning convolutional neural networks for graphs* [Paper presentation]. International Conference on Machine Learning, New York City, NY, USA.
- Song, C., Lin, Y. F., Guo, S. N., & Wan, H. Y. (2020). Spatial-Temporal Synchronous Graph Convolutional Networks: A New Framework For Spatial-Temporal Network Data Forecasting. *Proceedings of the AAAI Conference on Artificial Intelligence*, 34(01), 914–921. <https://doi.org/10.1609/aaai.v34i01.5438>
- Shuman, D. I., Narang, S. K., Frossard, P., Ortega, A., & Vandergheynst, P. (2013). The emerging field of signal processing on graphs: Extending high-dimensional data analysis to networks and other irregular domains. *IEEE Signal Processing Magazine*, 30(3), 83–98. <https://doi.org/10.1109/MSP.2012.2235192>
- Tang, Q., Yang, M., & Yang, Y. (2019). ST-LSTM: A deep learning approach combined spatio-temporal features for short-term forecast in rail transit. *Journal of Advanced Transportation*, 2019, 1–8. <https://doi.org/10.1155/2019/8392592>
- Wu, C. J., Schreiter, T., Horowitz, R., & Gomes, G. (2014). Traffic flow prediction using optimal autoregressive moving average with exogenous input-based predictors. *Transportation Research Record: Journal of the Transportation Research Board*, 2421(1), 125–132. <https://doi.org/10.3141/2421-14>
- Wu, Z. H., Pan, S. R., Long, G. D., Jiang, J., Zhang, C. Q. (2019, August 10–16). Graph WaveNet for Deep Spatial-Temporal Graph Modeling. *Proceedings of the Twenty-Eighth International Joint Conference on Artificial Intelligence*.
- Xu, K., Li, C., Tian, Y., Sonobe, T., Kawarabayashi, K. I., & Jegelka, S. (2018, July 10–15). *Representation learning on graphs with jumping knowledge networks* [Paper presentation]. International Conference on Machine Learning, Stockholm, SWE, Sweden.
- Xu, B., Shen, H., Cao, Q., Qiu, Y., & Cheng, X. (2019). *Graph Wavelet Neural Network* [Paper Presentation]. International Conference on Learning Representations, May 6–9, New Orleans, LA, USA.
- Xu, R., Xiang, H., Tu, Z., Xia, X., Yang, M. H., & Ma, J. (2022, October 24–28). V2X-ViT: Vehicle-to-Everything Cooperative Perception with Vision Transformer, ECCV, Tel Aviv TLV, Israel.
- Yu, Q., He, N., Ye, W., & Pang, L. (2021). *Research on the frequency prediction of barrier-free resources in airport terminal based on ARIMA model* [Paper presentation]. 2021 IEEE International Conference on Power Electronics, Computer Applications (ICPECA), January 22–24. [Paper presentation] Shenyang, LN, China. <https://doi.org/10.1109/ICPECA51329.2021.9362603>
- Yu, B., Yin, H., & Zhu, Z. (2017, August 19–25). *Spatio-temporal graph convolutional networks: A deep learning framework for traffic forecasting* [Paper presentation]. International Joint Conference on Artificial Intelligence, Melbourne, VIC, Australia
- Zhang, J., Chen, F., Cui, Z., Guo, Y., & Zhu, Y. (2021). Deep learning architecture for short-term passenger flow forecasting in urban rail transit. *IEEE Transactions on Intelligent Transportation Systems*, 22(11), 7004–7014. <https://doi.org/10.1109/TITS.2020.3000761>
- Zhang, J., Zheng, Y., & Qi, D. (2017). Deep spatio-temporal residual networks for citywide crowd flows prediction. *Thirty-first Proceedings of the AAAI Conference on Artificial Intelligence*, 31(1), 1655–1661. <https://doi.org/10.1609/aaai.v31i1.10735>